# Bowen Gu

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## RESEARCH INTERESTS

Wireless communications; green Internet of Things; backscatter communications; mobile edge computing; reconfigurable wireless systems; integrated sensing and communications.

## EDUCATION & APPOINTMENT

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| 2024-present | Xinjiang University, School of Computer Science and Technology - A-category Introduced Talent |
| 2021-2024 | Ph.D., Electronic Information Technology, Macau University of Science and Technology |
| 2018-2021 | M.Eng., Electronics and Communication Engineering, Chongqing University of Posts and Telecommunications |

## RESEARCH PROJECTS

**2027-2030** **PI**, NSFC Young Scientists Fund: Reliable Access and Transmission for Backscatter-Assisted Hybrid Active-Passive Communication Systems.

**2025-2028** **PI**, Xinjiang Natural Science Foundation Young Scientists Fund: User Cooperation Mechanism Optimization for Hybrid Active-Passive Communications in Green IoT.

**2026-2030** Participant, NSFC Major Program: Information Theory and Controllable Coding of Data Elements.

**Ongoing** Participant, Central Government-Guided Local Science and Technology Development Project: Edge Data Governance Innovation Platform Based on Intelligent Signal Detection and Processing.

## SELECTED JOURNAL PUBLICATIONS

1. H. Qin, **B. Gu (corresponding author)**, et al., “Fairness-Aware Computation Offloading in Wireless-Powered MEC Systems with Cooperative Energy Recycling,” *IEEE Transactions on Mobile Computing*, to be published, 2026.

2. H. Xie, D. Li, **B. Gu**, X. Yu, Y. Xu, and C. Tellambura, “Movable Antenna-Aided Wireless Systems: Concurrent or Cumulative Movement?,” *IEEE Transactions on Communications*, vol. 74, pp. 8829-8842, 2026.

3. **B. Gu**, D. Li, H. Xie, K. Yu, Q. Guan, and Y. Xu, “Exploring Hybrid Active and Passive Multiple Access via Slotted ALOHA-Driven Backscatter Communications,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 5, pp. 3317-3332, Oct. 2025.

4. **B. Gu**, D. Li, H. Ding, G. Wang, and C. Tellambura, “Breaking the Interference and Fading Gridlock in Backscatter Communications: State-of-the-Art, Design Challenges, and Future Directions,” *IEEE Communications Surveys & Tutorials*, vol. 27, no. 2, pp. 870-911, Apr. 2025.

5. H. Xie, D. Li, and **B. Gu**, “Enhancing Spectrum Sensing via Reconfigurable Intelligent Surfaces: Passive or Active Sensing and How Many Reflecting Elements are Needed?,” *IEEE Transactions on Wireless Communications*, 2024.

6. H. Xie, D. Li, and **B. Gu**, “Exploring Hybrid Active-Passive RIS-Aided MEC Systems: From the Mode-Switching Perspective,” *IEEE Transactions on Wireless Communications*, 2024.

7. **B. Gu**, D. Li, and H. Xie, “Computation-Efficient Backscatter-Blessed MEC With User Reciprocity,” *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 9026-9031, Jun. 2024.

8. Y. Xu, **B. Gu (corresponding author)**, Z. Gao, D. Li, Q. Wu, and C. Yuen, “Applying RIS in Multi-User SWIPT-WPCN Systems: A Robust and Environmentally-Friendly Design,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 1, pp. 209-222, Feb. 2024.

9. **B. Gu**, D. Li, Y. Liu, and Y. Xu, “Exploiting Constructive Interference for Backscatter Communication Systems,” *IEEE Transactions on Communications*, vol. 71, no. 7, pp. 4344-4359, Jul. 2023.

10. **B. Gu**, D. Li, Y. Xu, C. Li, and S. Sun, “Many a Little Makes a Mickle: Probing Backscattering Energy Recycling for Backscatter Communications,” *IEEE Transactions on Vehicular Technology*, vol. 72, no. 1, pp. 1343-1348, Jan. 2023.

11. **B. Gu**, H. Xie, and D. Li, “Act Before Another Is Aware: Safeguarding Backscatter Systems With Covert Communications,” *IEEE Wireless Communications Letters*, vol. 12, no. 6, pp. 1106-1110, Jun. 2023.

12. H. Xie, **B. Gu**, D. Li, Z. Lin, and Y. Xu, “Gain Without Pain: Recycling Reflected Energy From Wireless-Powered RIS-Aided Communications,” *IEEE Internet of Things Journal*, vol. 10, no. 15, pp. 13264-13280, Aug. 2023.

13. Y. Xu, **B. Gu (corresponding author)**, D. Li, Z. Yang, C. Huang, and K.-K. Wong, “Resource Allocation for Secure SWIPT-Enabled D2D Communications With alpha Fairness,” *IEEE Transactions on Vehicular Technology*, vol. 71, no. 1, pp. 1101-1106, Jan. 2022.

14. Y. Xu, **B. Gu**, R. Q. Hu, D. Li, and H. Zhang, “Joint Computation Offloading and Radio Resource Allocation in MEC-Based Wireless-Powered Backscatter Communication Networks,” *IEEE Transactions on Vehicular Technology*, vol. 70, no. 6, pp. 6200-6205, Jun. 2021.

15. Y. Xu, **B. Gu**, and D. Li, “Robust Energy-Efficient Optimization for Secure Wireless-Powered Backscatter Communications With a Non-Linear EH Model,” *IEEE Communications Letters*, vol. 25, no. 10, pp. 3209-3213, Oct. 2021.

## CONFERENCE PAPERS

1. H. Qin, **B. Gu (corresponding author)**, et al., “Lending a Hand: Cooperative Active-Passive Uplink Offloading for Low-Power MEC Networks,” *IEEE ICC Workshops*, 2026.

2. H. Qin, **B. Gu (corresponding author)**, X. Yu, H. Xie, Y. Xu, Q. Li, and L. Wang, “Revisiting Wireless-Powered MEC: A Cooperative Energy Recycling Framework for Task-Energy Co-Design,” *ICCT*, 2025.

3. X. Yu, D. Li, **B. Gu**, X. Jing, W. Wu, T. Wu, and K. Yu, “Meta-Learning Driven Lightweight Phase Shift Compression for IRS-Assisted Wireless Systems,” *IEEE GLOBECOM*, 2025.

4. **B. Gu**, Y. Xu, C. Huang, and R. Q. Hu, “Energy-Efficient Resource Allocation for OFDMA-Based Wireless-Powered Backscatter Communications,” *IEEE ICC*, 2021.

## HONORS & ACADEMIC SERVICE

2022: Outstanding Master's Thesis Award, Chongqing.   2021: Outstanding Master's Thesis Award, CQUPT.   2021-2024: PGS Full Scholarship, Macau University of Science and Technology.

Youth Editorial Board member for communications/information journals; TPC Member for WCSP 2023/2024 and IEEE WCNC 2026; IEEE and IEEE ComSoc Member; reviewer for IEEE JSAC, TWC, TMC, TCOM, IoTJ, TVT, TCCN, TGCN, WCL, and other journals.

## PATENTS

Four granted invention patents covering full-duplex relay MEC resource allocation, heterogeneous SWIPT resource allocation, robust allocation under imperfect channels, and energy-efficient wireless-powered backscatter networks.